DATA SCIENCE 301
DR. THEOBOLD

IMPACTS OF WEATHER CONDITIONS AND ROAD INFRASTRUCTURES ON CAR ACCIDENT SEVERITY IN THE STATE OF CALIFORNIA

JADERIAN REBOSURA, JAVIER ZIALCITA, JACKSON NELSON

CAL POLY

INTRODUCTION

Driving is built into our day-to-day lives, putting people of many demographics (age, driving experience, and attitudes) on the road, influencing driving behaviors and accident risk for all those involved (whether you are driving alone or carpooling). Culturally, more people are on the roads during rush hour commutes, holiday travel, and for nightlife activities, which affects how and when people drive and overall traffic conditions. Historical considerations, such as regional traffic density, quality of infrastructures, weather patterns, the way roads are designed, and even construction activity can also contribute to the likelihood of accidents. Additionally, much of our roadways are regulated, maintained, and policed by many organizations, law enforcement, traffic agencies, and governments. The dataset used to analyze these patterns takes traffic accident information from these sources, which have been collected by traffic sensors and cameras across 49 states from 2019-2023.

We aim to find which combination of weather conditions at certain traffic/road infrastructures increase the overall severity of a car accident in the state of California. As our secondary, we are also identifying which road infrastructure is associated with high severity accidents.

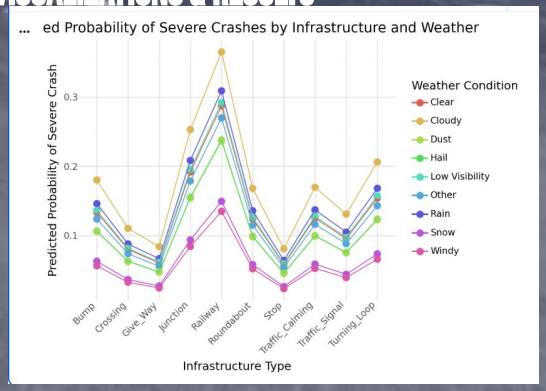
METHODS + MODELING

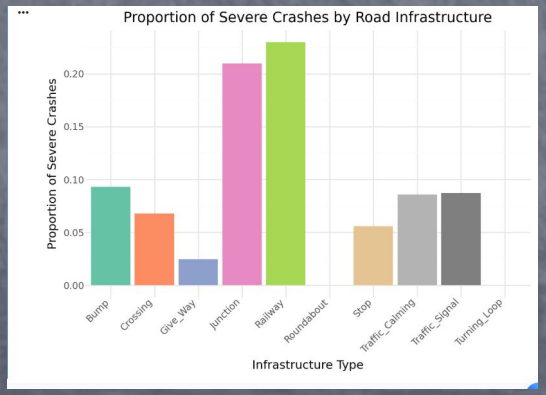
CLEANING:
1) replacing missing/ NAN values with zeros for missing numerical values, and a "missing" string for categorical values (important for calculations) **2)** "Weather_Condition" variable → subgroups to categorize each condition into: CLEAR, CLOUDY, RAIN, SNOW, LOW VISIBILITY, DUST, HAIL, and WINDY **3)** Separated features into categorical and numerical values **4)** "Severity" variable (1-4 scale) → into binary value for logistic regression (1-2 = less severe, 3-4 = more severe)

FEATURES:
Weather variables: Weather_Condition, Temperature, Visibility, Wind_Speed, Precipitation, Infrastructures: Bump, Crossing, Give_Way, Junction, Railway, Roundabout, Stop, Traffic_Calming, Traffic_Signal, Turning_Loop, State: California
MODELS: Logistic Regression, Cross-Validation, Accuracies

MODEL EXPLANATION:
To answer our primary question, we fit a logistic regression model to predict whether a car accident in California was severe. Predictor variables included the weather groupings, infrastructure indicators, and weather measurements. Weather categories were encoded using OneHotEncoder and quantitative variables were standardized using StandardScaler. Model performance was evaluated using 5-fold cross validation which produced an average accuracy of 0.83.
After fitting the model, we estimated the predicted probability of severe crashes across different weather and infrastructure combinations by creating hypothetical scenarios for each weather group and infrastructure type.
To address the secondary research question, we fit a second logistic regression model using only infrastructure variables. Model coefficients were extracted to determine which infrastructure types are most associated with severe crashes.

VISUALIZATIONS & RESULTS





CONCLUSION/ RESULTS

Primary Question: The model's results indicate that crashes occurring with railway infrastructure in the vicinity consistently produce the highest predicted probabilities of severe accidents across multiple weather conditions. For example, under cloudy conditions the predicted probability of a severe crash near a railway crossing is approximately 0.36, compared to lower probabilities for other infrastructure types. Similar patterns appear across several weather conditions, suggesting that the combination of railway infrastructure and adverse weather conditions may create particularly high-risk environments for severe crashes. This suggests that railway crossings may produce high risk areas for severe collisions. This could potentially be due to higher travel speeds, complex traffic interactions, or reduced reaction time near crossing signals.

Secondary Question: The results of the second model show that railway infrastructure is most strongly associated with severe accidents, having the largest positive coefficient of 0.8. Junctions are second, while stop signs and crossings are negative, suggesting they're associated with lower severity risk, likely because cars are more controlled here. These findings suggest that railway crossings may represent high-risk environments for severe crashes, while traffic control infrastructure may help reduce accident severity.

ETHICS/LIMITATIONS

One major ethical concern in regards to our model is that the results of the model could potentially be used to create harmful predictions. For example if our model determines that specific environment or infrastructure characteristics are associated with higher accident severity this could cause a number of potentially harmful outcomes. Insurance companies could use this data as justification to raise premiums or reduce services in areas that have these environmental conditions or types of infrastructure. These areas can also end up being subject to more intense policing and traffic monitoring. All of these things can also result in creating indirect bias by targeting areas that have these kinds of characteristics. This can make it so that communities that are already facing disadvantages end up being further penalized by this kind of data analysis.

One major limitation is that this data set was compiled using traffic sensors, public accident reports, and various transportation data sources. Because of this reliance accidents that occur in areas with few monitoring systems could very well end up being underreported. This means that the dataset may not be fully representing all accidents equally across the different locations creating potential coverage bias.

ACKNOWLEDGEMENTS

https://www.kaggle.com/datasets/sobhanmoosavi/us-accidents

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